CLINICAL · LARYNX & AIRWAY

Applied anatomy and physiology of the larynx

The larynx is a valve before it is a voice. It evolved to guard the lungs, closing in three tiers when we swallow and coughing out whatever gets past; phonation came later, borrowing the same folds. Its framework of cartilages, its membranes and spaces, and its two nerves explain nearly everything in laryngology: why glottic cancer rarely spreads to nodes, why a supraglottic one does, why a long left recurrent nerve is so often injured, and why a baby's airway closes so quickly.

AT A GLANCE

- Posterior cricoarytenoid is the only abductor.
- Cricothyroid is supplied by the external laryngeal nerve.
- The glottis has almost no lymphatics; supraglottis drains richly and bilaterally.
- In infants the subglottis is the narrowest part.

1 CHAPTER ONE Framework

Cartilages, joints and membranes.

1.1 Position

The larynx lies in front of the hypopharynx, opposite the third to sixth cervical vertebrae in the adult, ending at the lower border of C6. It rises and moves forwards in swallowing and can be moved from side to side over the spine with a grating feel, the *laryngeal crepitus*, which is lost when a postcricoid tumour fills the gap.

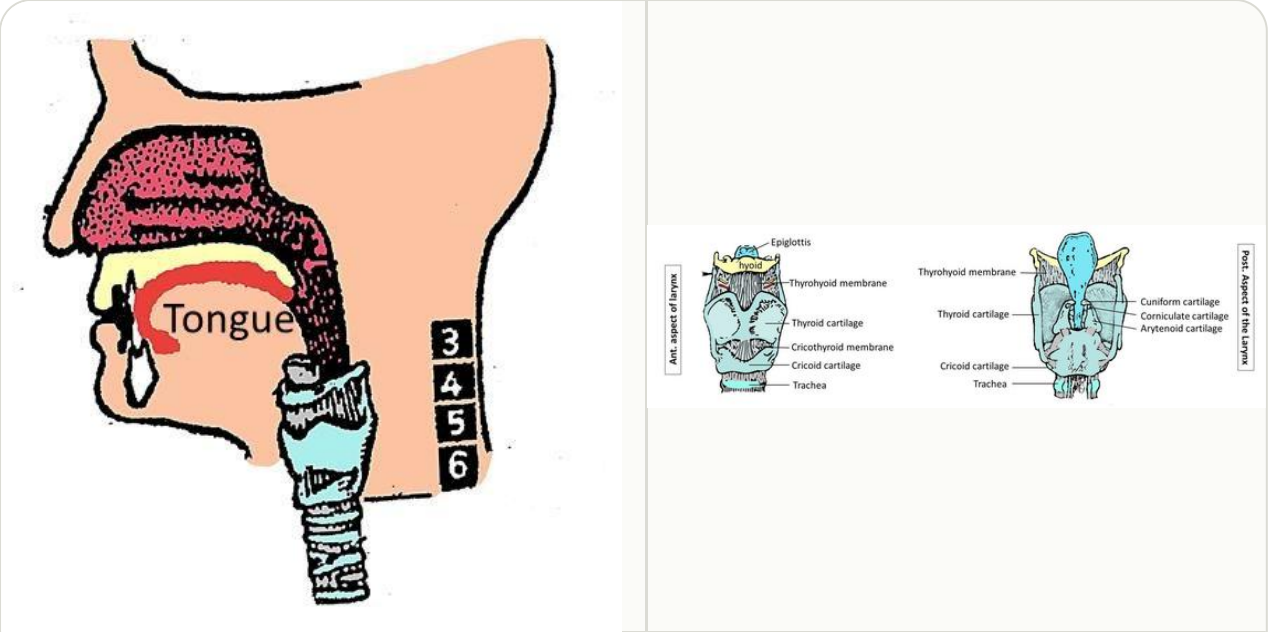


FIGURE 1 The site of the larynx from C3 to C6, and its anterior and posterior aspects.
Anatomy atlas · Doss p. 170

1.2 Cartilages

Three unpaired and three paired.

- **Thyroid:** the largest; two laminae meeting in front at about 90° in men (the Adam's apple) and 120° in women, with superior and inferior horns. The vocal folds attach to the inner surface of the angle at about its midpoint, so an airway can be made below them through the cricothyroid membrane.
- **Cricoid:** the only complete ring, a narrow arch in front and a tall lamina behind; it holds the airway open, and its loss or injury causes stenosis.
- **Epiglottis:** a leaf of elastic cartilage attached by its stalk (petiole) to the thyroid angle above the vocal folds, and by the hyoepiglottic ligament to the hyoid, which divides it into suprahyoid and infrahyoid parts. Its posterior bulge, the *tubercle*, hides the anterior commissure on mirror examination; pits and perforations in it let supraglottic cancer reach the pre-epiglottic space. It is not essential for swallowing and can be removed with little aspiration.
- **Arytenoids:** pyramids sitting on the cricoid lamina, each with a *vocal process* forwards (for the vocal ligament) and a *muscular process* laterally (for the cricoarytenoid muscles).
- **Corniculate (Santorini)** on the arytenoid apex, and **cuneiform (Wrisberg)** in the aryepiglottic folds.

Thyroid, cricoid and most of the arytenoid are hyaline and ossify from about 25 years; ossified cartilage can be mistaken for a foreign body on radiographs. The epiglottis, corniculate, cuneiform and arytenoid tips are elastic and do not ossify.

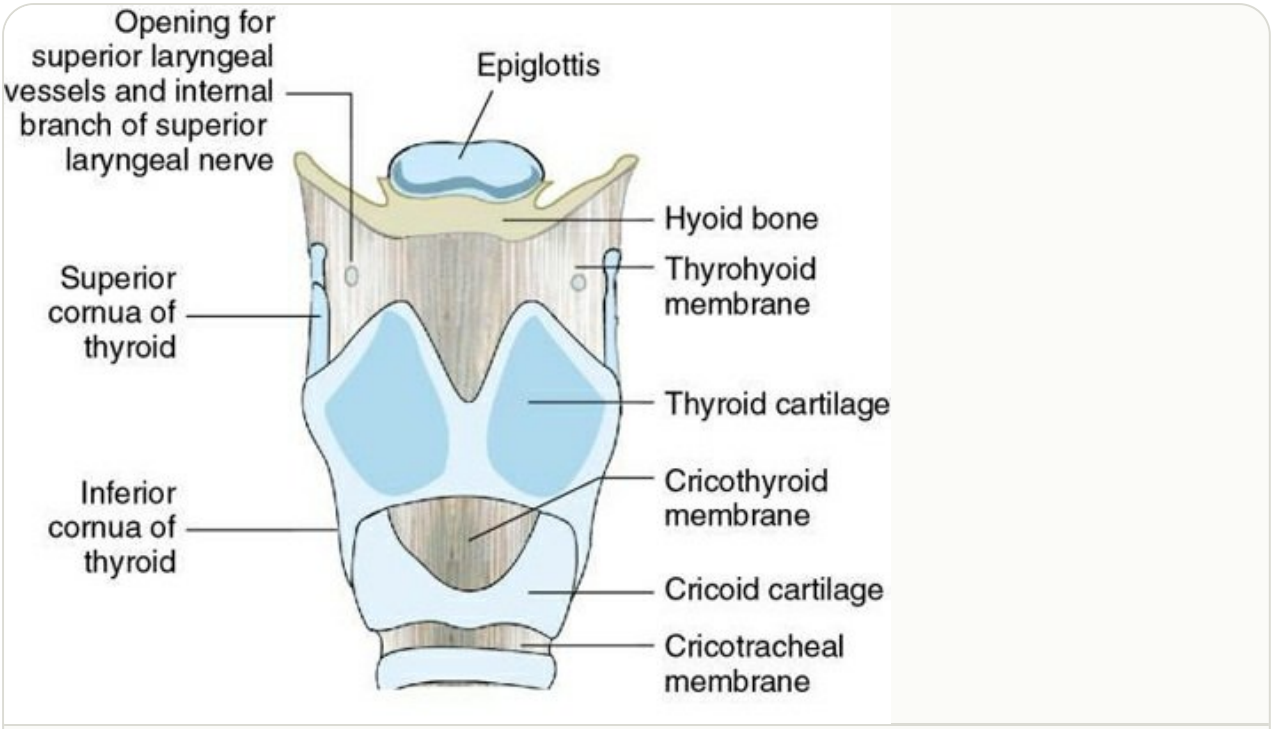


FIGURE 2 The laryngeal framework: thyroid, cricoid, epiglottis, hyoid and the thyrohyoid, cricothyroid and cricotracheal membranes.

Dhingra, fig. 56.1

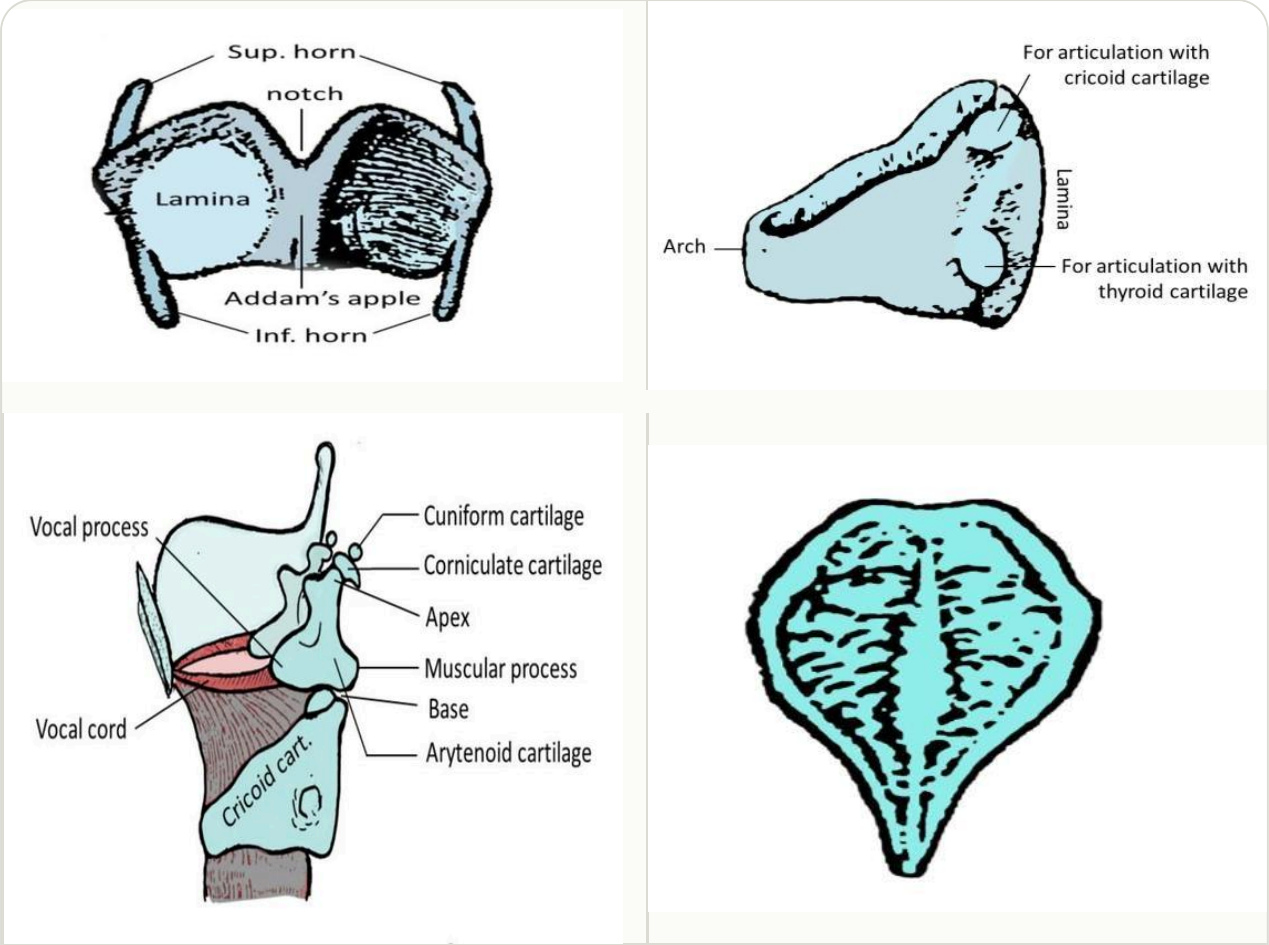


FIGURE 3 The thyroid, cricoid, arytenoid and epiglottic cartilages.

Anatomy atlas · Doss p. 171

1.3 Joints

- **Cricoarytenoid:** synovial; the arytenoid rotates about a vertical axis (abducting and adducting the vocal fold) and glides towards or away from its fellow (closing or opening the posterior glottis). Rheumatoid arthritis and intubation injury can fix it.
- **Cricothyroid:** synovial; the cricoid tilts on a transverse axis, lengthening and tensing the folds.

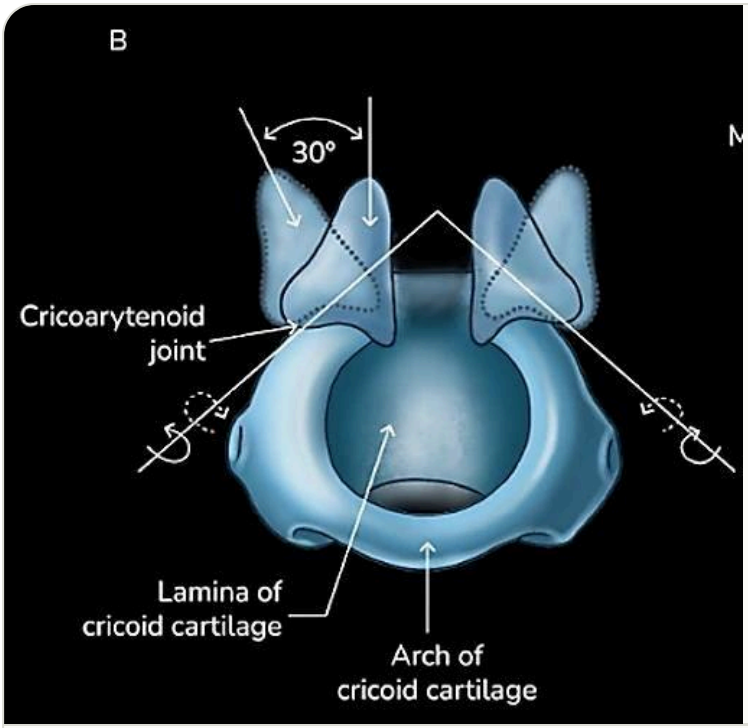


FIGURE 4 The cricoarytenoid joint.
Anatomy atlas · ENT Steps p. 234

1.4 Membranes and ligaments

- **Extrinsic:** the *thyrohyoid membrane* (pierced by the superior laryngeal vessels and internal laryngeal nerve), the cricotracheal membrane and the hyoepiglottic ligament.
- **Intrinsic:** the *cricovocal membrane (conus elasticus)*, whose free upper edge is the vocal ligament and whose two halves funnel up from the cricoid, a place where subglottic foreign bodies lodge; the *quadrangular membrane* between epiglottis and arytenoid, whose lower edge is the vestibular ligament; the *cricothyroid ligament* (the thickened front of the conus); and the thyroepiglottic ligament.

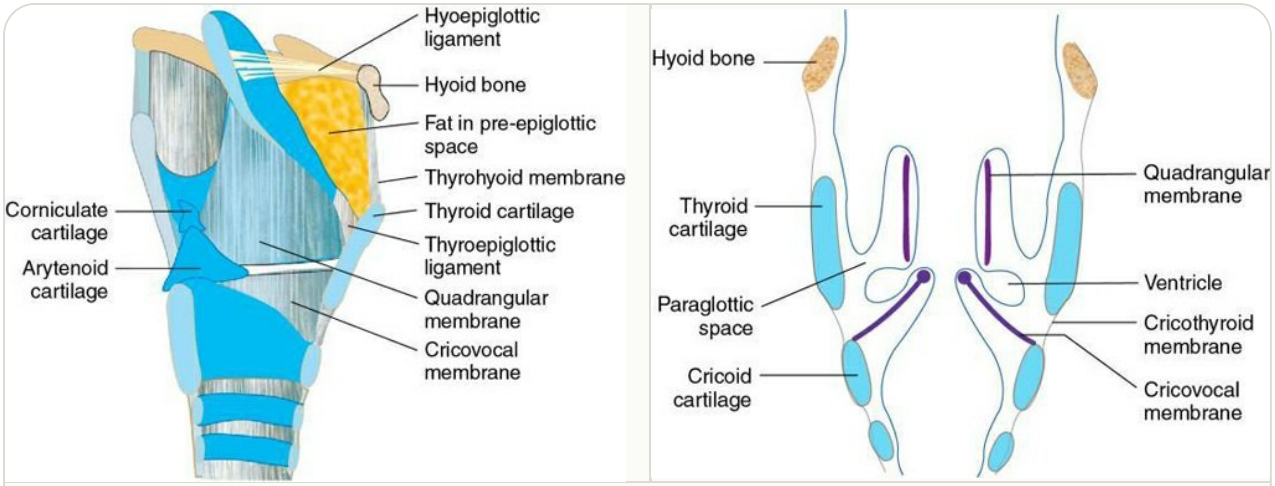


FIGURE 5 First, a sagittal section showing the cricovocal and quadrangular membranes and the pre-epiglottic space. Second, a coronal section: the quadrangular membrane ends in the false cord, the cricovocal membrane in the vocal ligament, forming the conus elasticus.

Dhingra, figs 56.2 and 56.3

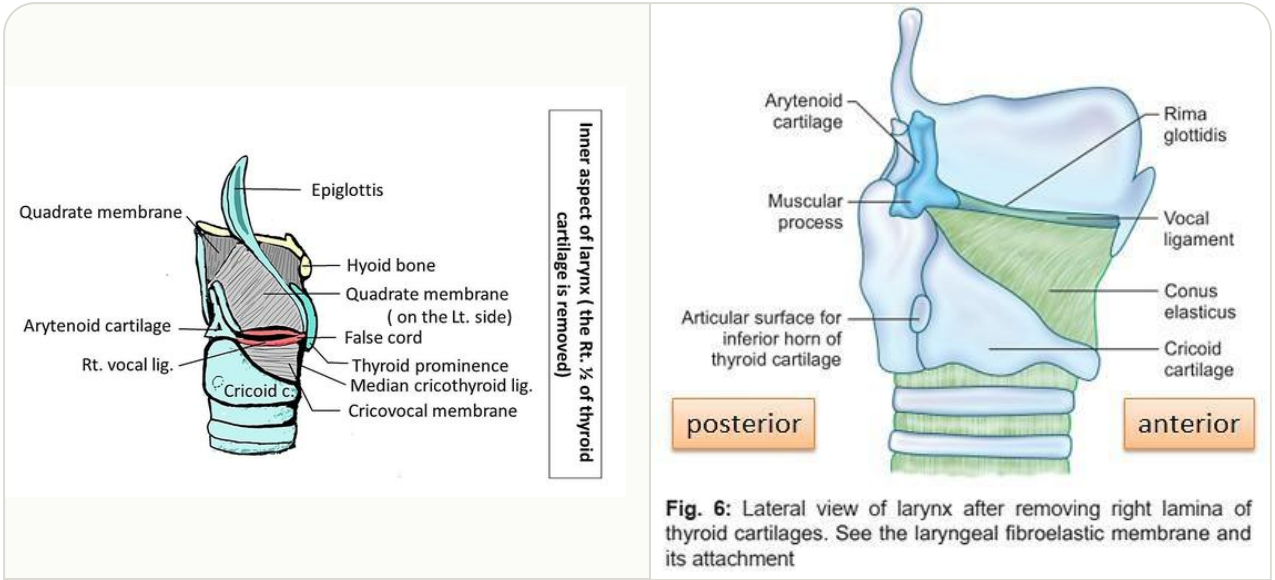


FIGURE 6 The membranes of the larynx, and the conus elasticus seen after removing a thyroid lamina.

Anatomy atlas · Doss p. 172, ENT Steps p. 237

2 CHAPTER TWO Muscles

Opening, closing, tensing.

2.1 Intrinsic muscles

INTRINSIC LARYNGEAL MUSCLES

ACTION	MUSCLES
Abduct the vocal folds	Posterior cricoarytenoid (the only abductor)
Adduct the vocal folds	Lateral cricoarytenoid, interarytenoid (transverse arytenoid), external thyroarytenoid
Tense the vocal folds	Cricothyroid (lengthens), vocalis (internal thyroarytenoid; fine tension)
Relax the vocal folds	Thyroarytenoid
Open the inlet	Thyroepiglottic
Close the inlet	Oblique interarytenoid and aryepiglottic

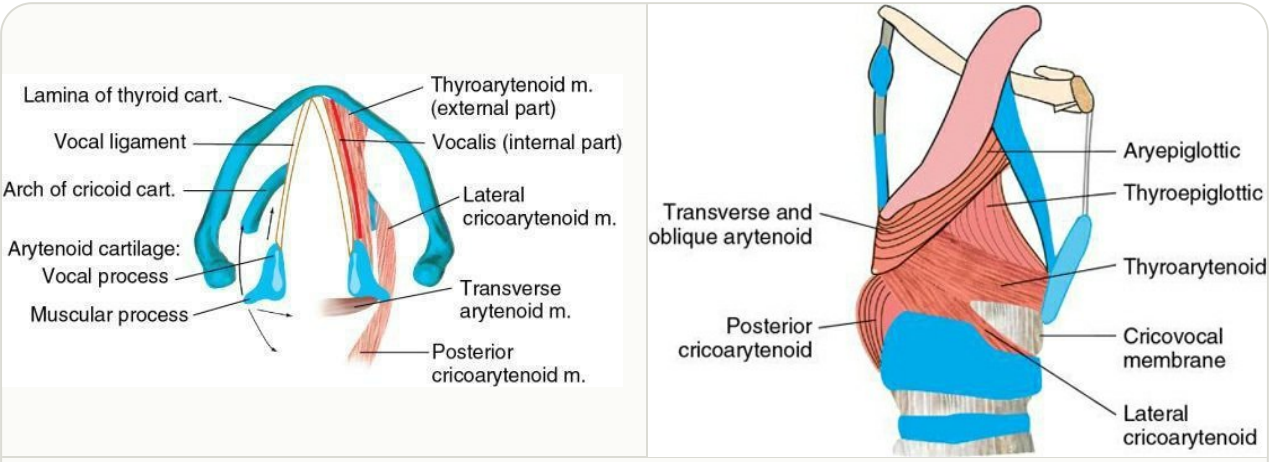


FIGURE 7 First, the laryngeal muscles and their actions on the arytenoid. Second, the intrinsic muscles seen from the side.

Dhingra, figs 56.4 and 56.5

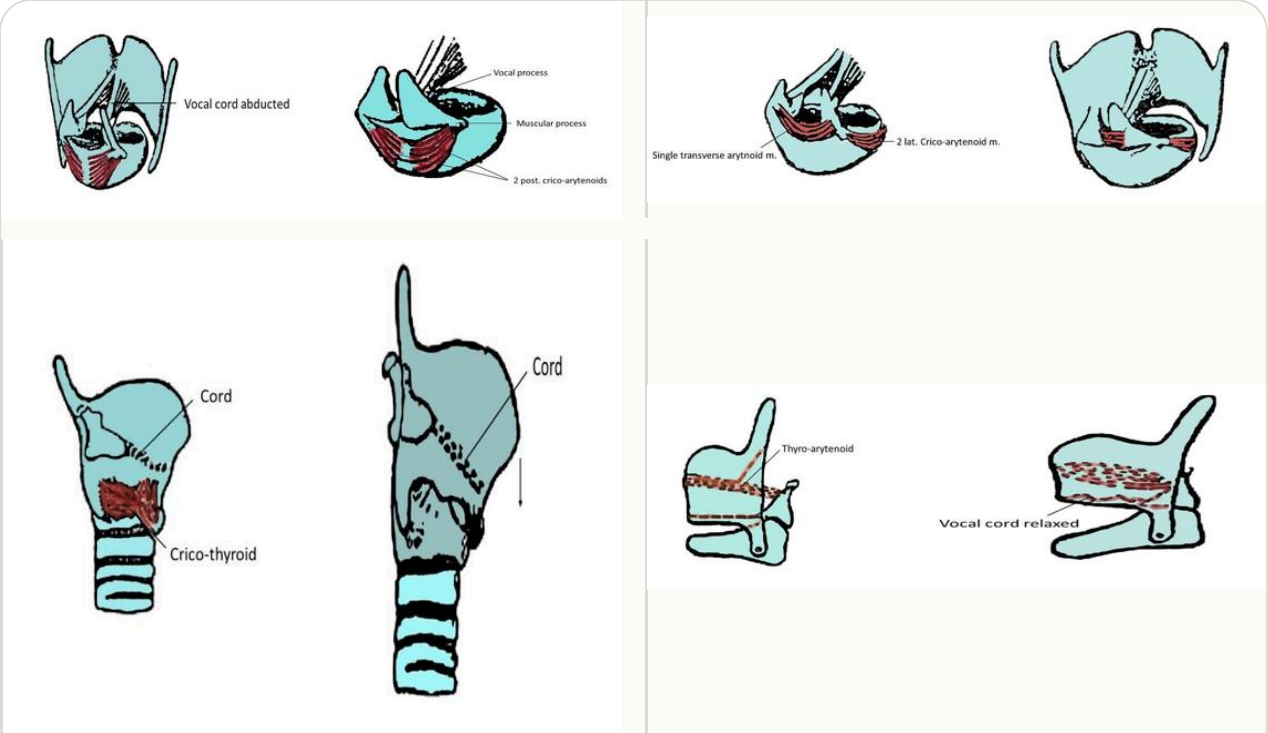


FIGURE 8 The posterior cricoarytenoid abducts; the lateral cricoarytenoid and interarytenoid adduct; the cricothyroid stretches and the thyroarytenoid relaxes the folds.
Anatomy atlas · Doss p. 174, Doss p. 173

THE ONE ABDUCTOR

The posterior cricoarytenoid alone opens the glottis. Lose it on both sides and the folds lie near the midline with a good voice and a poor airway.

2.2 Extrinsic muscles

- **Primary elevators** (attached to the thyroid cartilage): stylopharyngeus, salpingopharyngeus, palatopharyngeus and thyrohyoid.
- **Secondary elevators** (through the hyoid): mylohyoid, digastric, stylohyoid and geniohyoid.
- **Depressors:** sternohyoid, sternothyroid and omohyoid.

3 CHAPTER THREE Cavity and spaces

Above, at and below the folds.

3.1 The cavity

From the *inlet* (bounded by the epiglottis, aryepiglottic folds and interarytenoid fold) to the lower border of the cricoid. Two pairs of folds divide it:

- **Vestibule:** from the inlet down to the vestibular (false) folds.
- **Ventricle:** the recess between false and true folds; from its front a pouch, the *saccule*, rises between the false fold and the thyroid lamina, lubricating the folds with mucus. Its pathological enlargement is a *laryngocele*.
- **Subglottis (infraglottic space):** from the vocal folds to the lower border of the cricoid.

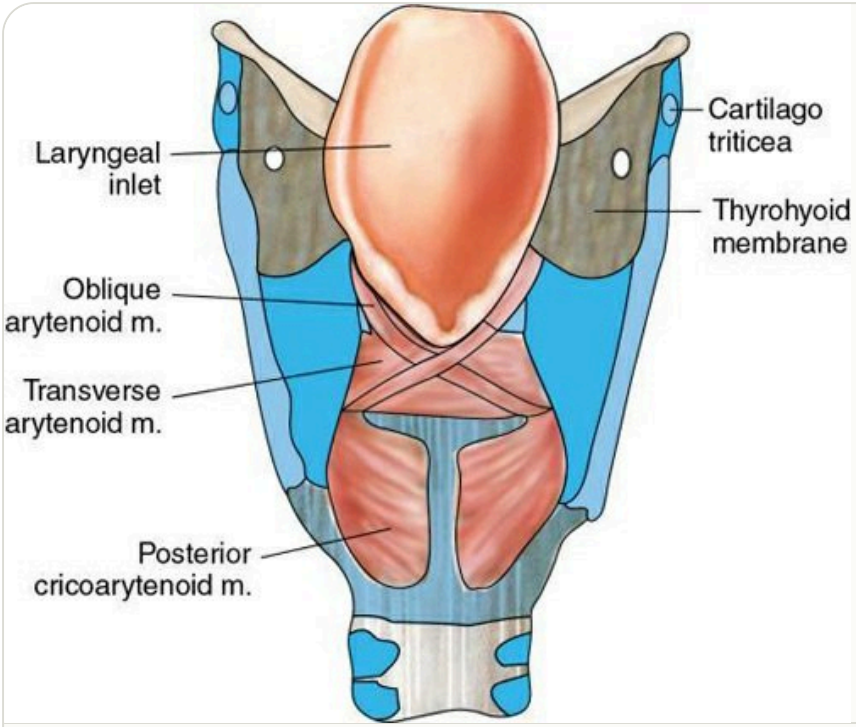


FIGURE 9 The laryngeal inlet and intrinsic muscles seen from behind.
Dhingra, fig. 56.6

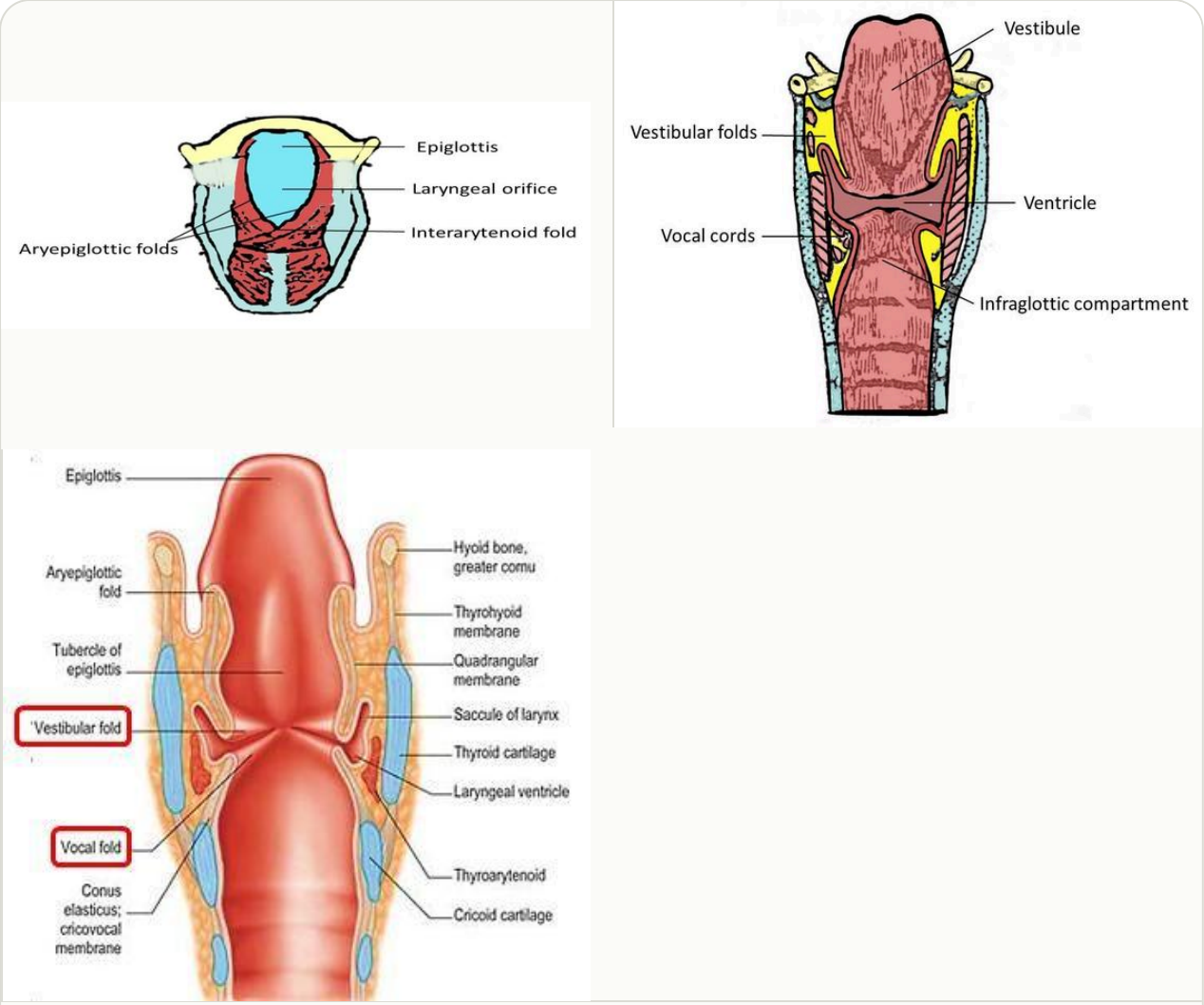


FIGURE 10 The laryngeal inlet, the cavity with its vestibule, ventricle and infraglottic part, and a coronal section with the saccule.

Anatomy atlas · Doss p. 175, Doss p. 174, ENT Steps p. 238

3.2 The vocal folds and glottis

The true folds run from the thyroid angle to the vocal processes: pearly white because they are covered by closely bound squamous epithelium without glands. The *glottis (rima glottidis)* is the gap between them, about 24 mm long in men and 16 mm in women, the narrowest part of the adult airway. Its anterior two-thirds, between the membranous folds, is the *phonatory glottis*; its posterior third, between the arytenoids, the *respiratory glottis*.

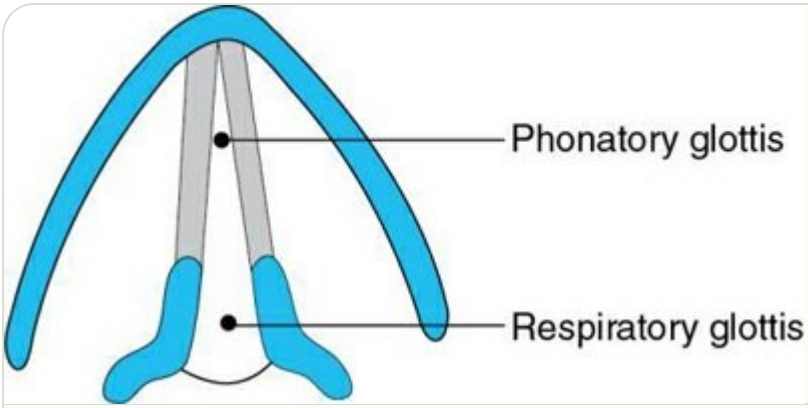


FIGURE 11 The rima glottidis: the anterior two-thirds is the phonatory glottis, the posterior third (between the arytenoids) the respiratory glottis.

Dhingra, fig. 56.7

3.3 Layers of the vocal fold

- Squamous epithelium.
- Lamina propria: a *superficial layer* (*Reinke's space*, loose and rich in hyaluronic acid), an intermediate layer (elastic) and a deep layer (collagen); the intermediate and deep layers form the *vocal ligament*.
- The vocalis muscle.

The epithelium and superficial lamina propria form the *cover*, which ripples over the stiffer *body* as the mucosal wave; scarring of the cover ruins the voice, and fluid in Reinke's space gives Reinke's oedema.

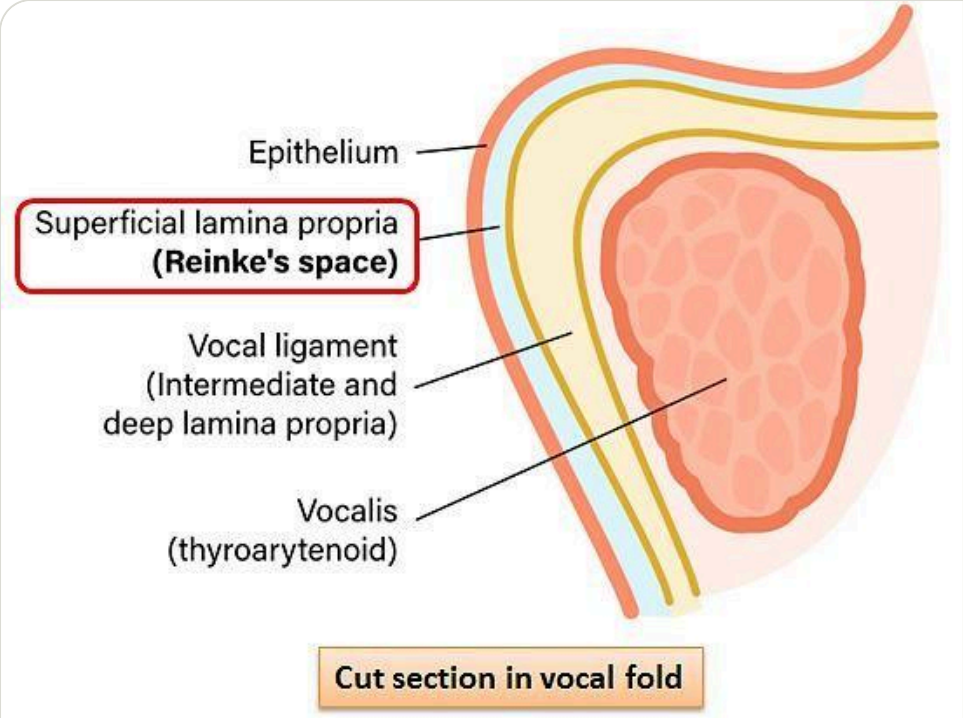


FIGURE 12 The layers of the vocal fold: Reinke's space and the vocal ligament.
Anatomy atlas · ENT Steps p. 246

3.4 Regions

- **Supraglottis:** epiglottis, aryepiglottic folds, arytenoids, false folds and ventricles.
- **Glottis:** the vocal folds with the anterior and posterior commissures, extending about 1 cm below the free edge.
- **Subglottis:** from there to the lower border of the cricoid.

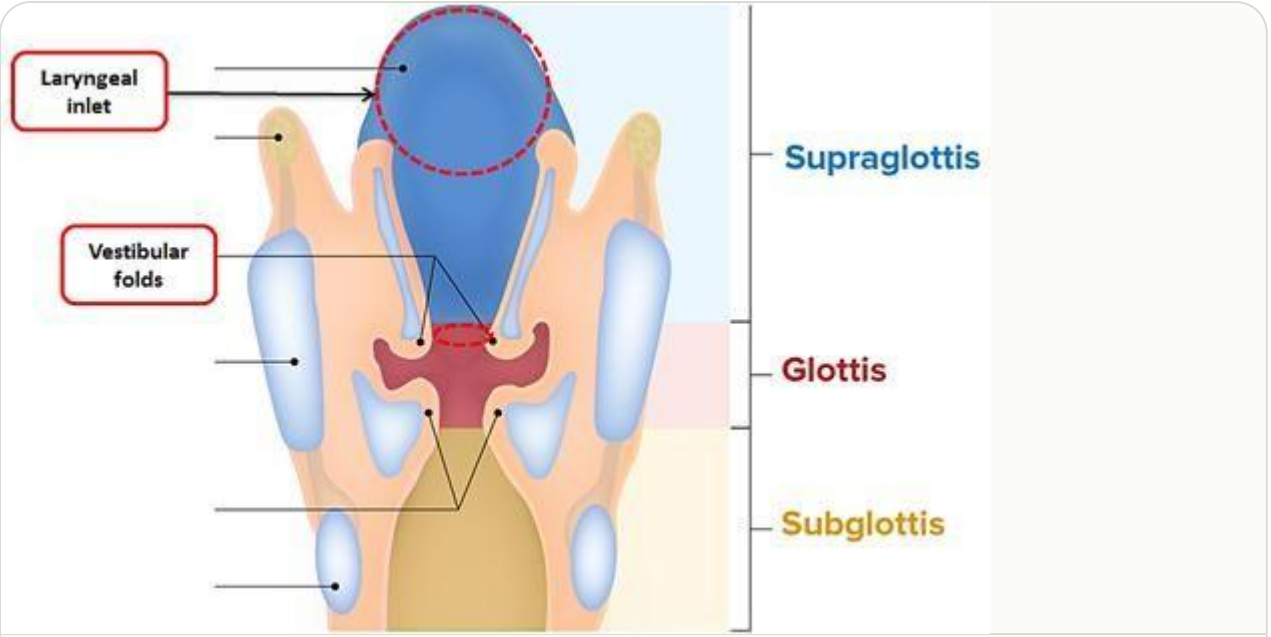


FIGURE 13 Supraglottis, glottis and subglottis.
Anatomy atlas · ENT Steps p. 239

3.5 Spaces

- **Pre-epiglottic space (of Boyer):** fat in front of the infrahyoid epiglottis, behind the thyrohyoid membrane and thyroid cartilage, below the hyoepiglottic ligament; invaded by supraglottic and tongue base cancers.
- **Paraglottic space:** between the thyroid lamina laterally and the ventricle, quadrangular membrane and conus elasticus medially, the piriform sinus behind; continuous with the pre-epiglottic space. Cancers spreading in it are transglottic and can leave the larynx through the cricothyroid membrane.
- **Reinke's space:** under the epithelium of the membranous fold, between the superior and inferior arcuate lines.

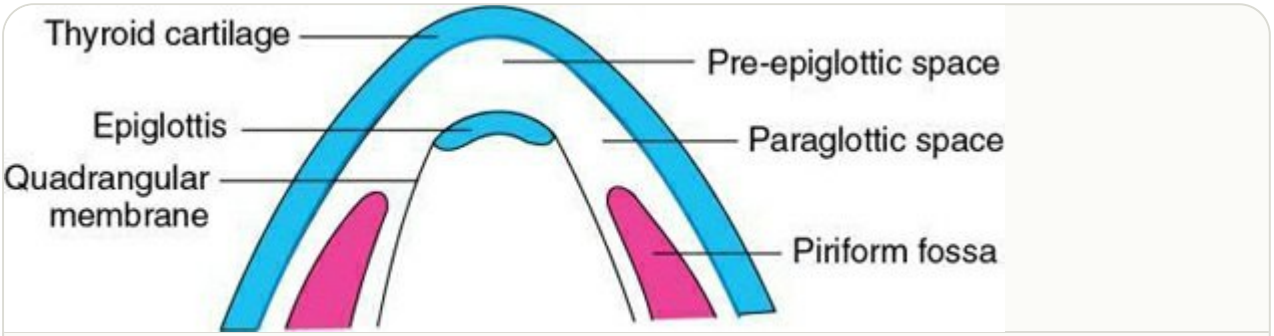


FIGURE 14 The paraglottic and pre-epiglottic spaces communicate with each other.
Dhingra, fig. 56.8

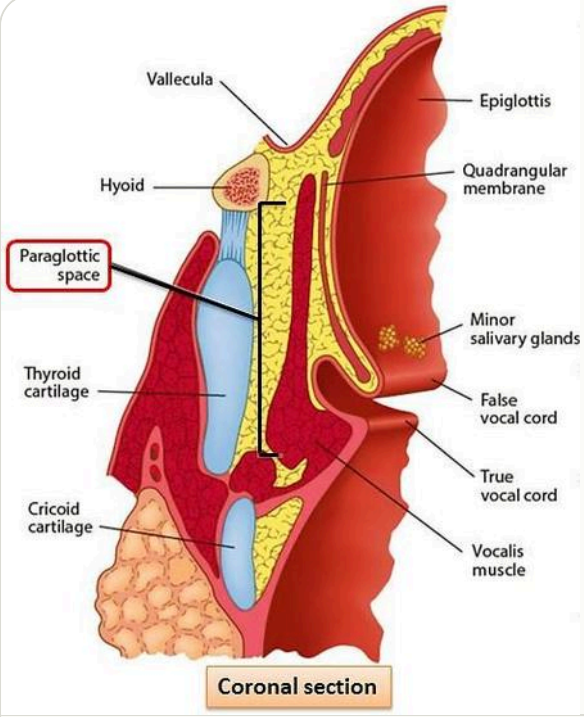


FIGURE 15 The paraglottic space in coronal section.
Anatomy atlas · ENT Steps p. 244

3.6 Mucosa

Ciliated columnar epithelium except over the vocal folds and upper vestibule (squamous). It is loosely attached, and swells readily, everywhere except over the posterior epiglottis, vocal folds and the corniculate and cuneiform cartilages. Glands are plentiful on the posterior epiglottis, aryepiglottic folds and saccule, and absent from the vocal folds.

4 CHAPTER FOUR Nerves, vessels and lymph

The supply.

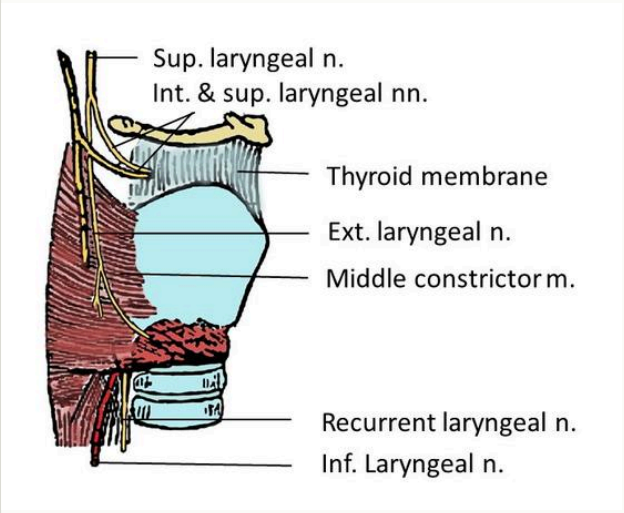
4.1 Superior laryngeal nerve

From the inferior (nodose) ganglion of the vagus, it descends behind the internal carotid artery and divides at the level of the greater horn of the hyoid:

- **Internal branch:** sensory to the mucosa above the vocal folds, piercing the thyrohyoid membrane with the superior laryngeal artery; it runs under the mucosa of the piriform fossa.
- **External branch:** motor to the *cricothyroid*, running with the superior thyroid artery, where it is at risk in thyroidectomy (loss of high pitch and vocal stamina).

4.2 Recurrent laryngeal nerve

Motor to every intrinsic muscle except the cricothyroid (the interarytenoid receives both sides), and sensory below the vocal folds. The *left* nerve leaves the vagus in the chest, hooks under the arch of the aorta beside the ligamentum arteriosum and climbs in the tracheo-oesophageal groove; the *right* hooks around the subclavian artery. Both enter the larynx behind the cricothyroid joint. The left, with its long course through the chest, is paralysed far more often. A *non-recurrent* right nerve accompanies an aberrant retro-oesophageal right subclavian artery.



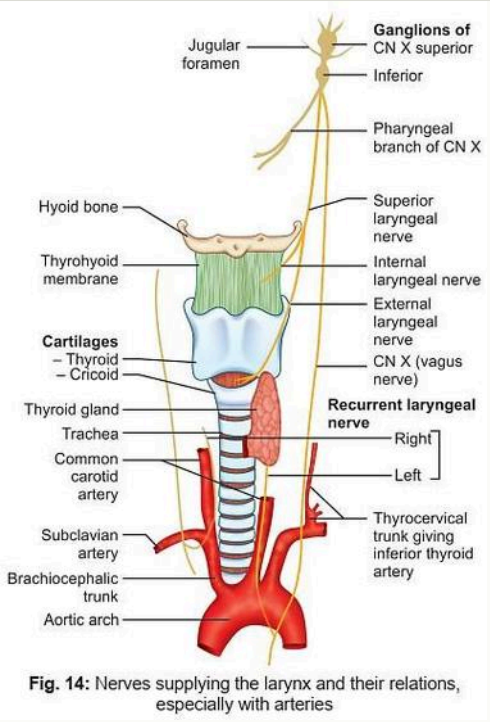


Fig. 14: Nerves supplying the larynx and their relations, especially with arteries

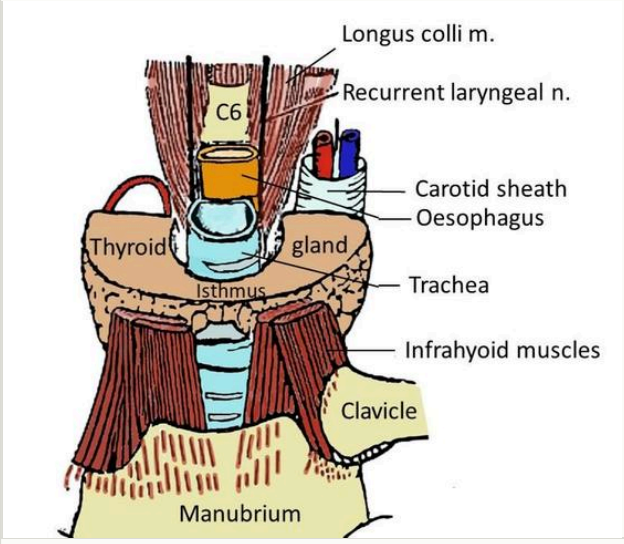


FIGURE 16 The nerve supply of the larynx, the nerves and their relations, and the cervical trachea with the recurrent laryngeal nerve.

Anatomy atlas · Doss p. 175, ENT Steps p. 251, Doss p. 176

4.3 Embryology

The larynx develops from the laryngotracheal groove of the foregut; the cartilages and muscles come from the fourth and sixth arch mesenchyme. The *superior laryngeal nerve* is the nerve of the fourth arch (hence the cricothyroid), and the *recurrent laryngeal nerve* that of the sixth (all other intrinsic muscles); the sixth arch artery on the left becomes the ductus arteriosus, which is why the left nerve hooks

around it. The lumen is obliterated in the sixth week and recanalises by the tenth: failure gives webs and atresia.

4.4 Vessels

The superior laryngeal artery (from the superior thyroid) with the internal laryngeal nerve, and the inferior laryngeal artery (from the inferior thyroid) with the recurrent nerve; matching veins drain to the internal jugular and brachiocephalic veins.

4.5 Lymphatic drainage

- **Supraglottis:** rich lymphatics through the thyrohyoid membrane to the upper and middle deep cervical nodes, often both sides.
- **Glottis:** almost none; hence early glottic cancer rarely metastasises.
- **Subglottis:** through the cricothyroid membrane to the prelaryngeal (Delphian), pretracheal and paratracheal nodes, and to the lower deep cervical and mediastinal nodes.

5 CHAPTER FIVE The child's larynx

Not a small adult larynx.

- **High:** opposite C3 or C4 at rest, rising to C1 or C2 in swallowing; the epiglottis meets the soft palate, so an infant can breathe through the nose while milk passes either side of the epiglottis. The larynx descends to C5 to C6 by adulthood.
- **Soft:** the cartilages collapse easily; the epiglottis is omega-shaped and the arytenoids are relatively large.
- **Overlapping:** the hyoid overlaps the thyroid and the thyroid overlaps the cricoid, so the cricothyroid and thyrohyoid membranes are narrow and hard to find; cricothyroidotomy is avoided in young children.
- **Funnel-shaped:** the *subglottis (cricoid ring)* is the narrowest part, about 4 to 5 mm in the newborn; one millimetre of oedema there halves the airway. A diameter under 4 mm in a term baby is subglottic stenosis.
- **Loose submucosa** swells quickly with infection and trauma.

The larynx grows rapidly in the first three years, which allows some congenital narrowings to be watched, and again at puberty, when the male vocal folds lengthen and the voice breaks. The folds measure about 6 to 8 mm in childhood and 15 to 23 mm in adults.

6 CHAPTER SIX Physiology

Four functions.

6.1 Protection of the lower airway

The oldest function. During a swallow three sphincters close in turn: the *inlet* (aryepiglottic folds, epiglottic tubercle and arytenoids), the *false folds*, and the *true folds*, the last barrier; breathing stops reflexly; the larynx rises and moves forward under the base of tongue as the epiglottis tilts back. Anything that gets through provokes a *cough*: the folds close, pressure builds below them, and a sudden release blows the material out. The larynx is the watchdog of the lungs.

6.2 Phonation

The *myoelastic–aerodynamic theory*: the adducted folds are blown apart by subglottic pressure from the lungs and fall back by their elasticity and the Bernoulli effect, releasing puffs of air that vibrate the cover as a mucosal wave. Loudness depends on subglottic pressure; pitch on the frequency of vibration, set by length, tension and mass of the folds. The pharynx, mouth, nose and chest resonate the sound, and the lips, tongue, palate and teeth articulate it into speech.

6.3 Respiration

The folds abduct on inspiration and partly adduct on expiration, regulating airflow and providing back pressure to the lungs.

6.4 Fixation of the chest

A closed glottis fixes the chest so that the arm, trunk and abdominal muscles can work against it in lifting, climbing, coughing, vomiting, defecation, micturition and

childbirth.

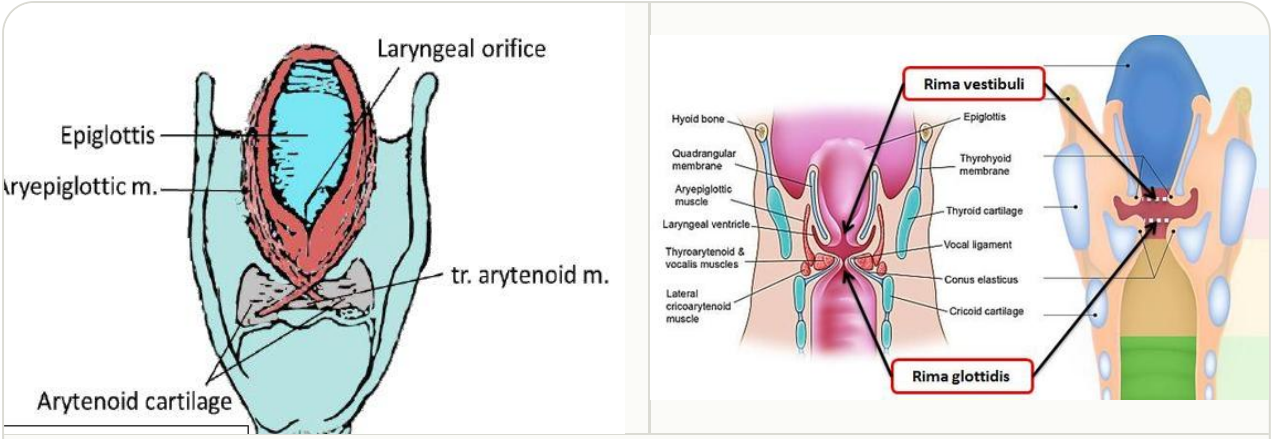


FIGURE 17 The muscles that close the laryngeal inlet, and the rima vestibuli and rima glottidis.
Anatomy atlas · Doss p. 173, ENT Steps p. 241

7 CHAPTER SEVEN For the examination

What the examiner wants to hear.

7.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Cartilages,** which are paired, and which ossify.
- **Intrinsic muscles** and their actions; the only abductor.
- **Conus elasticus,** quadrangular membrane, pre-epiglottic and paraglottic spaces.
- **Layers of the vocal fold** and Reinke's space.
- **Nerve supply** and the course of the recurrent laryngeal nerves.
- **Lymphatic drainage** by region and its importance in cancer.
- **The infant larynx.**
- **Functions** of the larynx.

7.2 Common pitfalls

COMMON PITFALLS

- Forgetting that the cricothyroid is supplied by the external laryngeal nerve.
- Calling the glottis the narrowest part of the infant airway; it is the subglottis.
- Mistaking ossified laryngeal cartilage for a foreign body on a radiograph.

Written for this app from Oxford Textbook of Otolaryngology ch. 47, Oxford Specialist Handbook ch. 8, Dhingra ch. 56, 60 and K. J. Lee ch. 50. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

LARYNX & TRACHEA · TUMOURS

Glottic Tumor

IMAGING FINDINGS

- Thickening or mass of the true cord
- Anterior commissure thicker than 1 mm
- Paraglottic and subglottic spread

TOP DIFFERENTIALS

- Benign cord lesion
- Chondrosarcoma (cricoid, chondroid matrix)
- Cord paralysis (medial arytenoid)

CHECKLIST

- Cord movement on breath-hold images
- Thyroid cartilage erosion: inner cortex T3, through it T4a



CT with intravenous contrast enhancement, axial. Slight asymmetry of the vocal cords can be observed with slight enlargement of the vocal…



Axial CECT shows an enhancing right TVC exophytic mass. Anterior and posterior commissures are normal. Right arytenoid cartilage sclerosis is…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 530.

LARYNX & TRACHEA · INFECTION & INFLAMMATION

Laryngocele

IMAGING FINDINGS

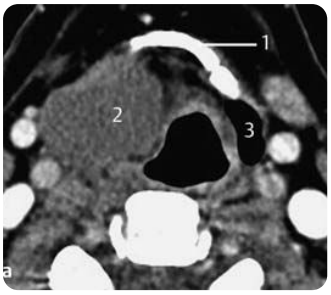
- Air or fluid-filled sac from the laryngeal ventricle
- Internal (paraglottic) or external through the thyrohyoid membrane
- Rim enhancement if infected (pyolaryngocele)

TOP DIFFERENTIALS

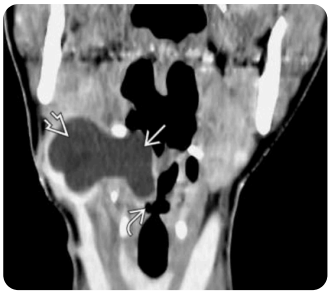
- Thyroglossal duct cyst
- Second branchial cleft cyst
- Saccular cyst

CHECKLIST

- Exclude a carcinoma at the ventricle in adults
- Airway compromise



CT with intravenous contrast enhancement, axial. The axial slice at the lower border of the hyoid (1) reveals a homogeneous, expanding…



Coronal CECT reformat shows a similar lesion. This is also known as a "mixed" laryngocele because it contains internal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 292.